Tutorial of ADMET prediction of drugs and molecular properties calculation by MolAICal

Qifeng Bai
Email: molaical@yeah.net
Homepage: https://molaical.github.io/baiqf.html
School of Basic Medical Sciences
Lanzhou University
Lanzhou, Gansu 730000, P. R. China

1. Introduction

About 40% of drug candidates have failed in the past due to toxicity and unacceptable efficacy. In this case, the predictions of drug ADMET (**absorption, distribution, metabolism, excretion, and toxicity**) play an important role in the success of a drug candidate. In this tutorial, MolAICal is introduced to calculate ADMET and molecular properties of drugs by implement the interface package of FP-ADMET [1] tool and improved adme-pred-py under the permitted GNU General Public License v3.0 and AGPL-3.0 license. For more detailed MolAICal, please visit: https://molaical.github.io.

2. Materials

2.1. Software requirements

1) MolAICal: https://molaical.github.io or https://molaical.gitlab.io

Notice: Make sure the installation of MolAICal is correct!

2.2. Example files

1) All the necessary tutorial files are downloaded from:
https://github.com/MolAICal/tutorials/tree/master/018-admet

Part I. ADMET calculation of drugs

1. Install FP-ADMET model

Currently, only **Linux version** of MolAICal is supported in Part I tutorial.

1) Click and open the website: **DownloadModel**

Then go to the folder "AImodels→ADMET→FPADMET→linux64" in the open website, download a file named "fpadmet.tar.gz"

2) Move "fpadmet.tar.gz" to the folder "MolAICal-xxx/mtools". Where "MolAICal-xxx" is your decompressed root directory of MolAICal. The folder "mtools" is the targeted directory.

3) Decompress file

#> tar -xzvf fpadmet.tar.gz

And go to folder named fpadmet:

#> cd fpadmet

4) install fpadmet model

#> chmod +x install.sh

#> ./install.sh

Until now, fpadmet model is installed completely.

2. Calculate ADMET of drugs

2.1. Input file format for ADMET calculation

Users should prepare the right format for the input file. One molecule occupies one line of the input file. For one line, the first column contains molecular SMILES string, the second column includes the molecular name. And the molecular name should have one space with molecular SMILES string at least (see Figure 1).

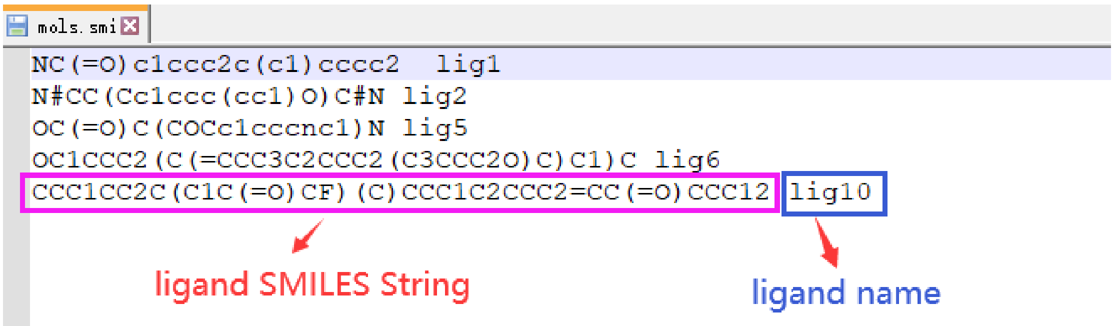


Figure 1. Content of input file

2.2. Calculate all items of ADMET from a file containing molecular SMILES

Go to 018-admet, and use the below command:

#> molaical.exe -model fpadmet -s all -i mols.smi -o results.dat

Interpretations of results: It will generate a file named "results.dat" that contains all items of ADMET of drug. Open file "results.dat", it will show 58 item results for one drug. For example, predicting Blood–brain-barrier penetration (item 2) and pKa dissociation constant (item 50) are shown below :

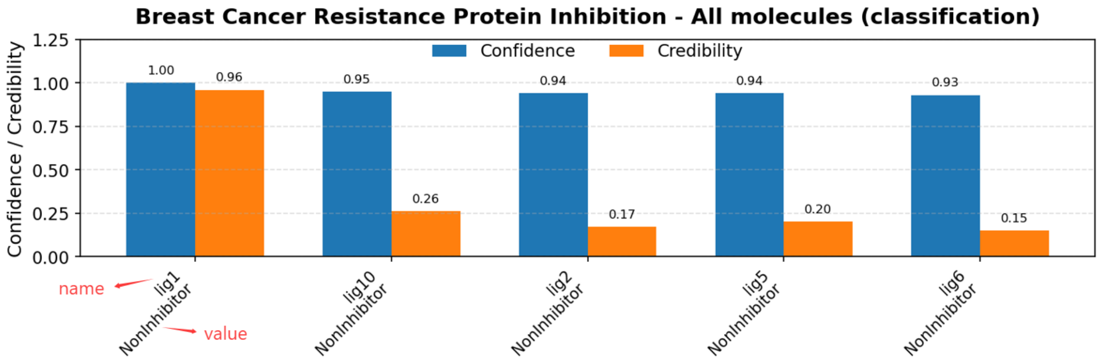
```
################################
2: Blood-brain-barrier penetration
Predicted Confidence Credibility
lig1 Yes 0.93 0.32

…………………………………………….

50: pKa dissociation constant
Predicted quantile_0.025 quantile_0.975
lig1 5.19 -1.60 13.06
################################
```

"lig1" is a drug or ligand name. The label "Yes" of compound "lig1" indicates that this compound is predicted to be suitable for blood brain barrier penetration. A confidence value of 0.93 indicates that the classifier is quite certain, and the prediction is possible to be a single label. A relatively low value of credibility (0.32) suggests that the compounds like "lig1" are not sufficiently represented in the training set, and users should deal with the prediction with caution. For the regression example of item 50, a 95% prediction interval (predictions at 0.025 and 97.5 percentiles for pKa) is computed and gives a range for the predictions on individual observation. As the original paper on FP-ADMET said: narrow prediction intervals indicate a lower uncertainty associated with the prediction [1]. For more detailed interpretations of predicted results, please check the files named "ADMETModels-doc.pdf", "FP-ADMET.pdf" and "FP-ADMET-SI.pdf" in folder "doc" along with this tutorial document.

To visualize the results, the following command can either generate plots of 58 property values for each individual molecule (where each molecule has its own folder named after it), or generate images for each property entry across all molecules, which are saved in the folder "merge_lig":
#> molaical.exe -call run -c sfile -i m_admetp.py -i results.dat --merge

It will generate plots of 58 property values for each individual molecule in the folders that have the same names as the ligand names (e.g., lig1, lig2…). And generate images for each property entry across all molecules in the folder "merge_lig". **The results are provided as PNG images and PDF documents, with the results in the PDF documents being consistent with the image results.**

The example figure is shown below. The x-axis represents the molecular names and their corresponding ADMET values. Please refer to the calculation results when interpreting this figure:



2.3. Calculating a specified item of ADMET based on a file that contains SMILES strings
Go to 018-admet, and use the below command:
#> molaical.exe -model fpadmet -s single -i mols.smi -n 1

Interpretations of results: It will generate a file named "1.dat" due to selecting task item 1. If task item 2 is chosen, it will produce a file named "2.dat". If task item 3 is chosen, it will produce a file named "3.dat"… So the prefix name of generated result file is the same as the selected task item. Open file "1.dat", the content of "1.dat" is like below:

```
##############################
Predicted Confidence Credibility
lig1 Negative 0.77 0.31
```

lig2 Negative 0.90 0.59
lig5 Negative 0.80 0.35
lig6 Negative 0.91 0.64
lig10 Negative 0.96 0.86
################################

It only shows the calculated results of "1: Anticommensal Effect on Human Gut Microbiota" item of ADMET for all molecules. For understanding all items of ADMET, please check the MolAICal manual, Appendix in this tutorial, or the files named **"ADMETModels-doc.pdf"** in the folder doc, "FP-ADMET.pdf" and "FP-ADMET-SI.pdf" in folder "doc" along with this tutorial document.

To visualize the results, it can find the figure according to the item name in the above-generated folder "merge_lig".

Part II. Calculations of ADME and molecular properties by MolAICal

In this part, **Linux or Windows version** of MolAICal is supported for this part tutorial.

In this part tutorial, it will show the usages of MolAICal for the calculations of ADME and molecular properties.

1) If "-s" is 0, it will show all calculated molecular properties information
#> molaical.exe -model admepro -s 0 -i "O=C(C)Oc1ccccc1C(=O)O"

2) If "-s" is 1, it will show the brief general ADME information of a ligand
#> molaical.exe -model admepro -s 1 -i "O=C(C)Oc1ccccc1C(=O)O"

For example, the output is shown below:
1. Pharmacokinetics:
GI absorption: High
BBB permeant: Yes

2. Lipinski Rule of 5 Violations:
No violations found

3. Medicinal Chemistry:
PAINS filter: False
Brenk filter: True

3) If "-s" is 2, it will show basic molecular properties, for example: molecular weight, Number of Hydrogen Bond Donors, etc.
#> molaical.exe -model admepro -s 2 -i "O=C(C)Oc1ccccc1C(=O)O"

For more detailed items of molecular properties, please check MolAICal manual.

Reference

1. Venkatraman V. FP-ADMET: a compendium of fingerprint-based ADMET prediction models. J Cheminform. 2021;13(1):75.

Appendix

58 ADMET items of FP-ADMET are listed as follows:

1: Anticommensal Effect on Human Gut Microbiota

2: Blood-brain-barrier penetration

3: Oral Bioavailability

4: AMES Mutagenecity

5: Metabolic Stability

6: Rat Acute LD50

7: Drug-Induced Liver Inhibition

8: HERG Cardiotoxicity

9: Haemolytic Toxicity

10: Myelotoxicity

11: Urinary Toxicity

12: Human Intestinal Absorption

13: Hepatic Steatosis

14: Breast Cancer Resistance Protein Inhibition

15: Drug-Induced Choleostasis

16: Human multidrug and toxin extrusion Inhibition

17: Toxic Myopathy

18: Phospholipidosis

19: Human Bile Salt Export Pump Inhibition

20: Organic anion transporting polypeptide 1B1 binding

21: Organic anion transporting polypeptide 1B3 binding

22: Organic anion transporting polypeptide 2B1 binding

23: Phototoxicity human

24: Phototoxicity in vitro

25: Respiratory Toxicity

26: P-glycoprotein Inhibition

27: P-glycoprotein Substrate

28: Mitochondrial Toxicity

29: Carcinogenecity

30: DMSO Solubility

31: Human Liver Microsomal Stability

32: Human Plasma Protein Binding

33: hERG Liability

34: Organic Cation Transporter 2 Inhibition

35: Drug-induced Ototoxicity

36: Rhabdomyolysis

37: T1/2 Human

38: T1/2 Mouse

39: T1/2 Rat

40: Cytotoxicity HepG2 cell line

41: Cytotoxicity NIH 3T3 cell line

42: Cytotoxicity HEK 293 cell line

43: Cytotoxicity CRL-7250 cell line

44: Cytotoxicity HaCat cell line

45: CYP450 1A2 Inhibition

46: CYP450 2C19 Inhibition

47: CYP450 2C9 Inhibition

48: CYP450 2D6 Inhibition

49: CYP450 3A4 Inhibition

50: pKa dissociation constant

51: logD Distribution coefficient (pH 7.4)

52: logS

53: Drug affinity to human serum albumin

54: MDCK permeability

55: 50% hemolytic dose

56: Skin penetration

57: CYP450 2C8 Inhibition

58: Aqueous Solubility (in phosphate saline buffer)